

TOBI OBADIAH
AI Engineer

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SUMMARY

AI Engineer with **6+** years of experience building backend systems, machine learning applications, and distributed cloud platforms. Experienced in Python, C++, FastAPI, AWS, and AI infrastructure, with a track record of designing scalable systems and leading engineering teams across fintech, AI, and blockchain products.

SKILLS

- **Expert:** Python, C++, C, FastAPI, Django, AWS, PostgreSQL, MySQL, Linux, Git, REST APIs
- **Professional:** TypeScript, Node.js, Docker, Kafka, Microservices, Distributed Systems, Graph Databases, RAG, Vector Databases, Deep Learning, LLMs, CI/CD, Blockchain

EXPERIENCE

Backend Engineer (contract) Neptune Technologies LLC 03/2024 - 10/2024

- Built **FastAPI** backend services for fintech CRM workflows, improving scalability and maintainability.
- Integrated Stripe subscriptions API into the CRM, enabling seamless automated billing and payment processing.
- Collaborated with cross-functional teams to streamline feature integration, reducing deployment time by 30%.

Full-stack Developer N01ZET 04/2022 - 05/2024

- Led a 5-engineer team, improving app performance by 60% with ReactJS, TypeScript, and Node.js.
- Created and deployed EOSIO blockchain smart contracts, enabling secure transactions for 10+ organizations.
- Implemented Kafka-based **microservices**, scaling to thousands of users and reducing response time by 30%.

Machine Learning Engineer Mboa Digital 11/2020 - 06/2021

- Developed ETL **machine learning pipeline** for automated bank statement and invoice reconciliation with 90% accuracy.
- Evaluated XGBoost, Random Forest, Linear Regression, SVMs, and neural networks for financial document matching.
- Integrated Google Document AI and deployed scalable inference pipelines on AWS Lambda and API Gateway.

Software and AI Engineer Artificialis & Engineering 11/2019 - 10/2020

- Engineered core features for the Ride & Smile transport application using **C++** and MySQL, optimizing performance.
- Optimized production deployment with comprehensive testing via Google's **GTest**, boosting feature validation by 50%.
- Developed Python and PySide (Qt) app for **data analysis** and visualization, improving data interpretation and reporting.

Freelance Software Developer Upwork - Fiverr 04/2018 - 05/2020

- Worked with clients across six continents in industries like transport and entertainment. Consistently delivered exceptional service, achieving **100% client satisfaction** and securing repeat business, reflected in **30+ five-star reviews**.

PROJECTS

Graph Database Engine

- A **High-performance** graph database in C with ACID transactions, WAL persistence, bitmap indexes, and vector search.
- Continuously evolving as an experimental platform for AI long-term memory research and production applications.
- Serves as the underlying storage engine for my MSc research on LLM-aware graph memory.

LLM-Aware Graph Memory Architecture (MSc Dissertation)

- Researching **graph-based long-term memory** architectures for LLM agents using semantic retrieval and graph traversal.
- Evaluating retrieval quality, token efficiency, scalability, and persistent knowledge storage
- Investigating graph databases as scalable long-term memory backends for persistent and context-aware AI agents.

Research Interests: AI Memory Systems (Semantic, Episodic & Procedural), Cognitive Architectures, Agentic AI, Reinforcement Learning, Knowledge Representation, Graph Databases.

EDUCATION

Masters of Science in Artificial Intelligence	De Montfort University	09/2025 - 09/2026
Bachelor of Science in Computer Science	University of Buea	10/2015 - 03/2019